Home Wi-Fi Network in Cisco Packet Tracer:::

Project file name: HomeWiFiNetwork.pkt

Objective

Create a network with three PCs and one wireless router, assign static IPv4 addresses, and verify connectivity using ping.

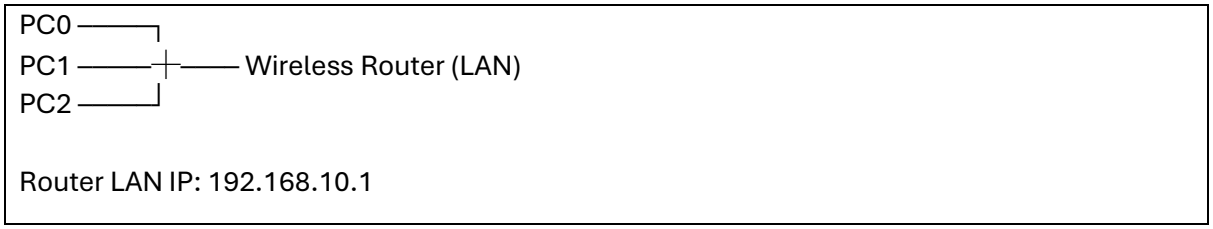
1. Devices and Connections

1 wireless router (e.g., WRT300N)

3 PCs (PC0, PC1, PC2)

3 Copper Straight-Through Ethernet cables

Suggested topology:



Place the router and three PCs in the workspace. Connect each PC's FastEthernet0 port to a LAN Ethernet port on the router. Do not use the WAN/Internet port for the PCs.

2. IP Addressing Plan

Device	IPv4 Address	Subnet Mask	Default Gateway
Router LAN	192.168.10.1	255.255.255.0	—
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.10.4	255.255.255.0	192.168.10.1

All devices use the 192.168.10.0/24 subnet. If the router's LAN address differs, configure it as 192.168.10.1 or update the PCs' gateway values to match.

3. Configure Each PC

PC0

1. Click PC0.
2. Open Desktop → IP Configuration.
3. Select Static.
4. IP Address: 192.168.10.2
5. Subnet Mask: 255.255.255.0
6. Default Gateway: 192.168.10.1

PC1

1. Click PC1.
2. Open Desktop → IP Configuration.
3. Select Static.
4. IP Address: 192.168.10.3
5. Subnet Mask: 255.255.255.0
6. Default Gateway: 192.168.10.1

PC2

1. Click PC2.
2. Open Desktop → IP Configuration.
3. Select Static.
4. IP Address: 192.168.10.4
5. Subnet Mask: 255.255.255.0
6. Default Gateway: 192.168.10.1

4. Save the Project

Select File → Save As and save the Packet Tracer project as HomeWiFiNetwork.pkt.

5. Test Connectivity

On each PC, open Desktop → Command Prompt and run these tests:

From PC0

ping 192.168.10.1
ping 192.168.10.3
ping 192.168.10.4

From PC1

ping 192.168.10.1
ping 192.168.10.2
ping 192.168.10.4

From PC2

ping 192.168.10.1
ping 192.168.10.2
ping 192.168.10.3

Expected result: replies from the destination IP. The first ping may time out while address resolution occurs; try again. If it fails, check cabling, link lights, IP addresses, masks, gateway, and router LAN configuration.

6. Results

Test	Expected Result	Actual Result
PC0 → PC1 (192.168.10.3)	Successful replies	Fill in after testing
PC0 → PC2 (192.168.10.4)	Successful replies	Fill in after testing
PC1 → Router (192.168.10.1)	Successful replies	Fill in after testing

PC2 → Router (192.168.10.1)	Successful replies	Fill in after testing

Conclusion

The network places three PCs and the router in the same IPv4 subnet. Static addressing and ping tests verify local connectivity. Add your own Packet Tracer screenshots and actual results before submission.